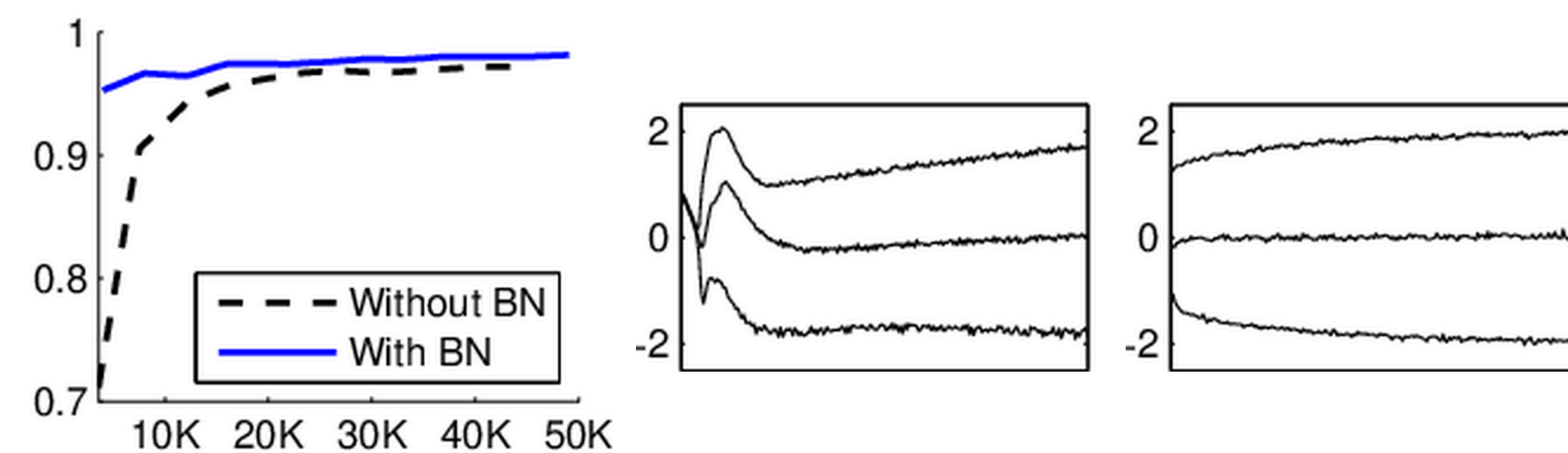
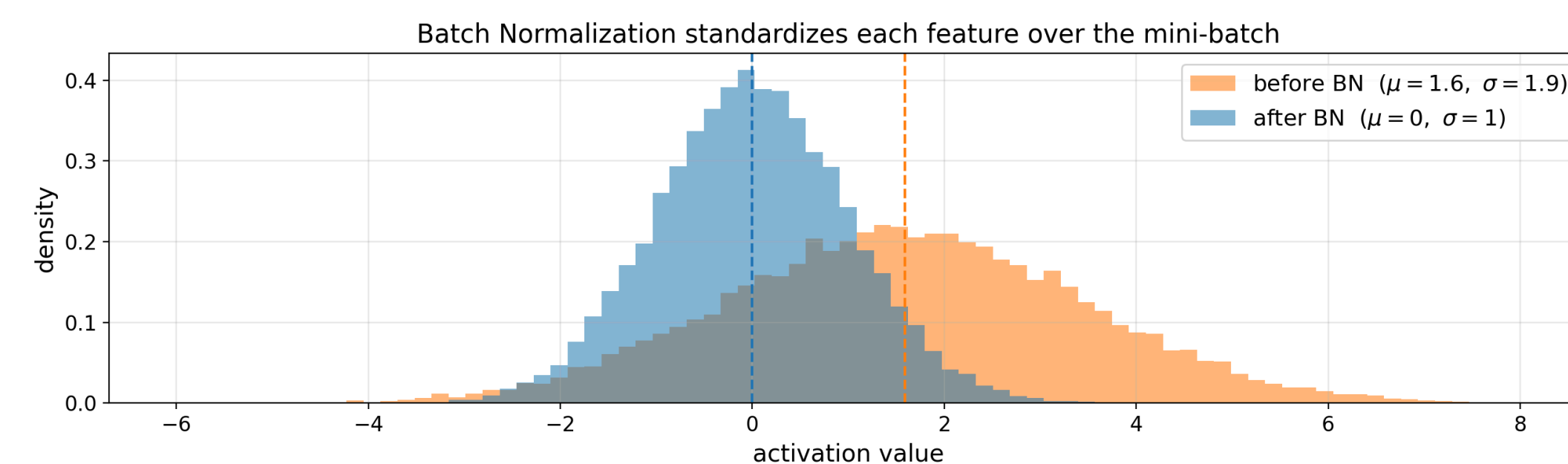


01. Introduction

Batch Normalization (BN) rescales every feature of a mini-batch to mean 0 and variance 1:

$$y = \gamma \cdot (x - \mu_B) / \sigma_B + \beta$$

This simple step lets networks train faster, reach **higher accuracy**, and keeps activation distributions **stable through training**, making BN a standard layer in deep learning. Its statistics come from the mini-batch itself, so its behavior depends on the **batch size**.



From Ioffe & Szegedy (2015): BN trains faster (left); an activation's percentiles during training, unstable without BN (middle), stable with BN (right).

02. Objective

Rank Reduction Autoencoders (Mounayer et al., 2024) regularize networks with the truncated **SVD**, no extra loss needed. Like BN, it **changes the statistics of the batch**.

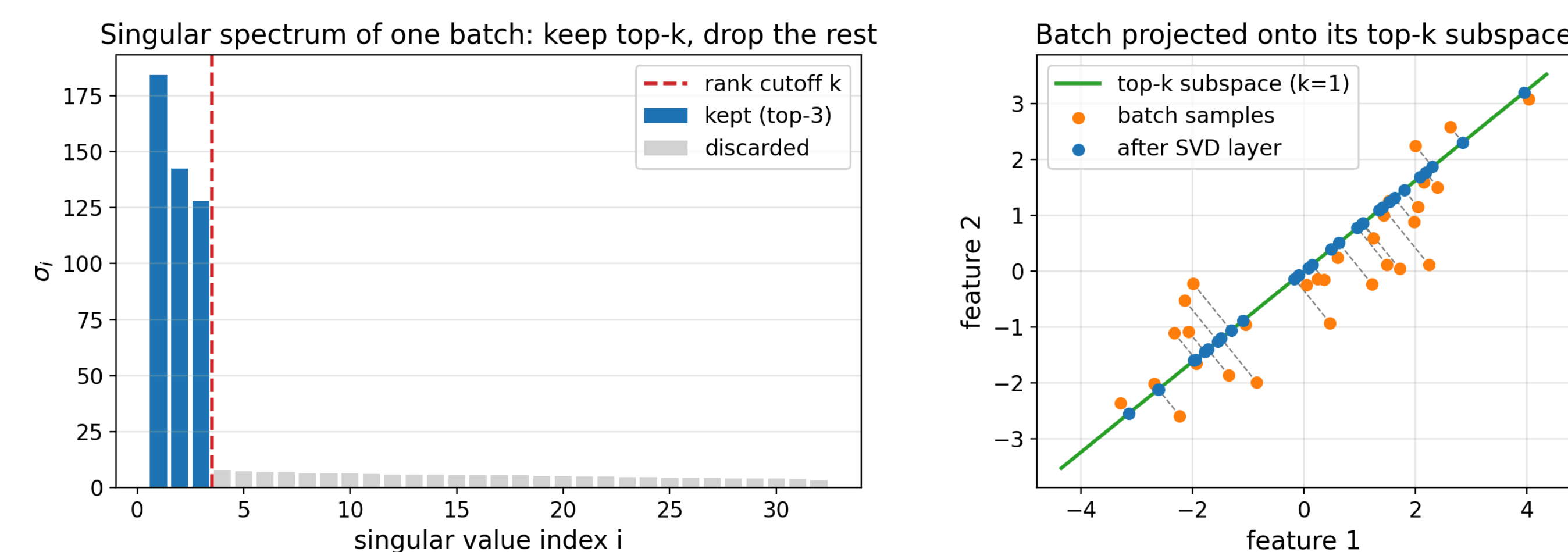
Can the SVD alone normalize a batch as well as BN? We build it into a standalone SVD layer to find out.

Setup at a glance: 64×64 images • 3 conv blocks (8→16→32 ch), no pooling • rank k = 8 • batch size 32 • Adam, lr 10⁻³ • 20 epochs (scaled up for small N) • up to 32 seeds per point • PyTorch

03. Methodology

The SVD layer rebuilds the batch from its **top-k singular components**, dropping everything else:

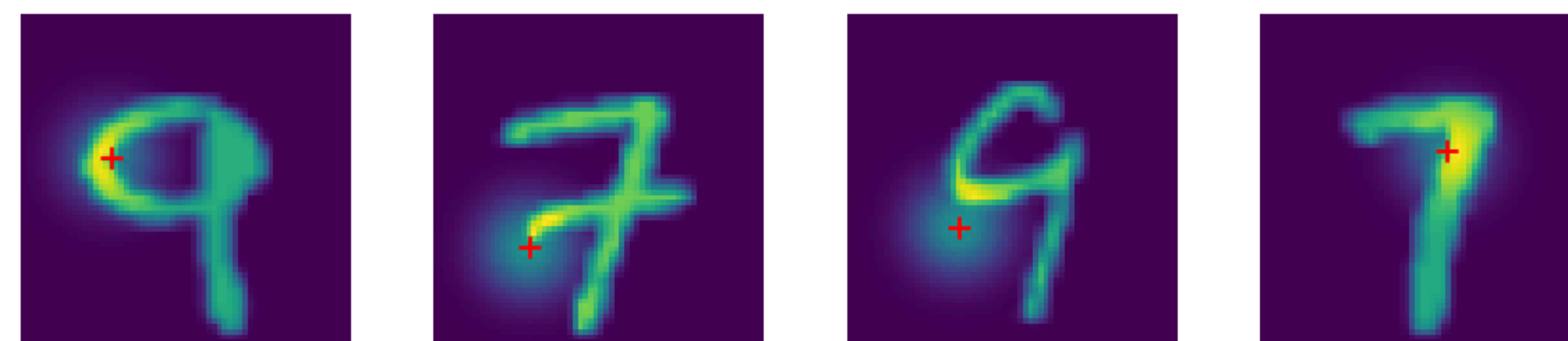
$$X = U \Sigma V^T \rightarrow \hat{X} = U_k \Sigma_k V_k^T$$



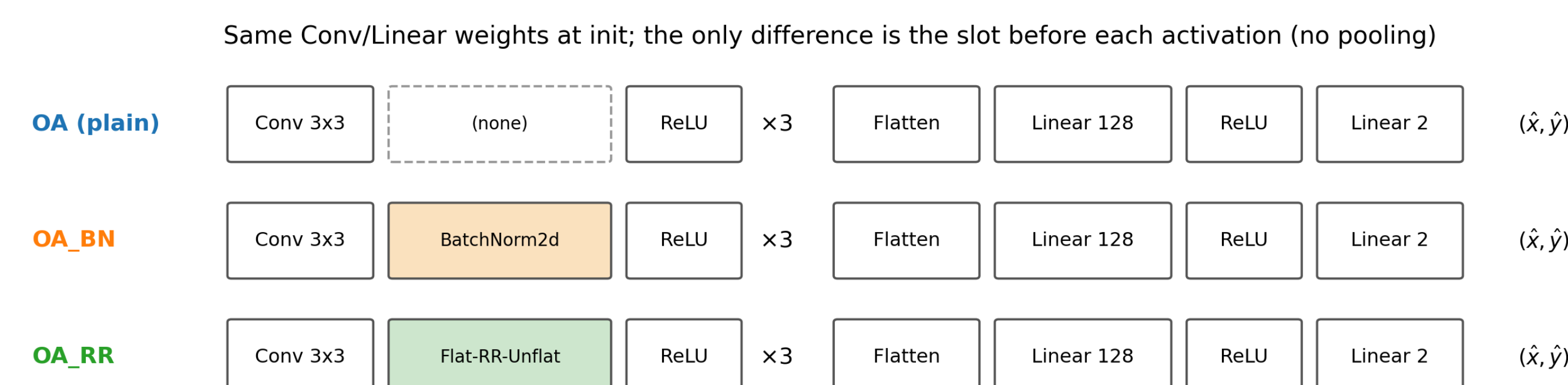
The batch, rebuilt from only its top-k singular components (right: k = 1 example).

- Task: regress a Gaussian source's centre (x, y) from a 64×64 image: **point-source localization**, as in astrometry and microscopy.
- Compare **Plain vs BN vs SVD** from **identical initial weights**, over up to 32 seeds, sweeping rank k, noise level and type, and dataset size N.
- External validity via **source injection**: a known source planted on real image backgrounds (astronomy's own validation method).

Injected source (red +) on real MNIST backgrounds

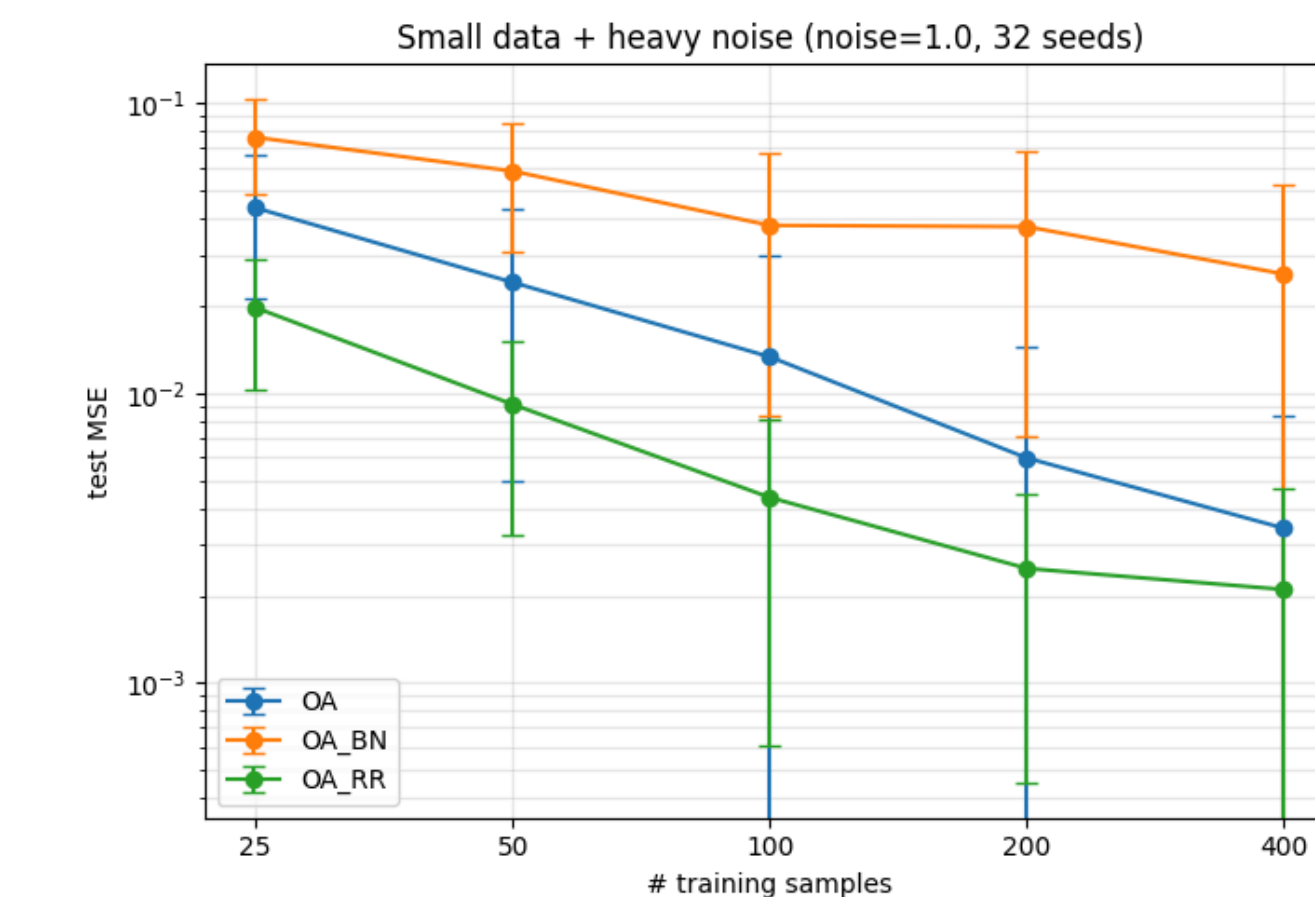


Injected source (red +) on real backgrounds: ground truth stays exact, clutter statistics become real.

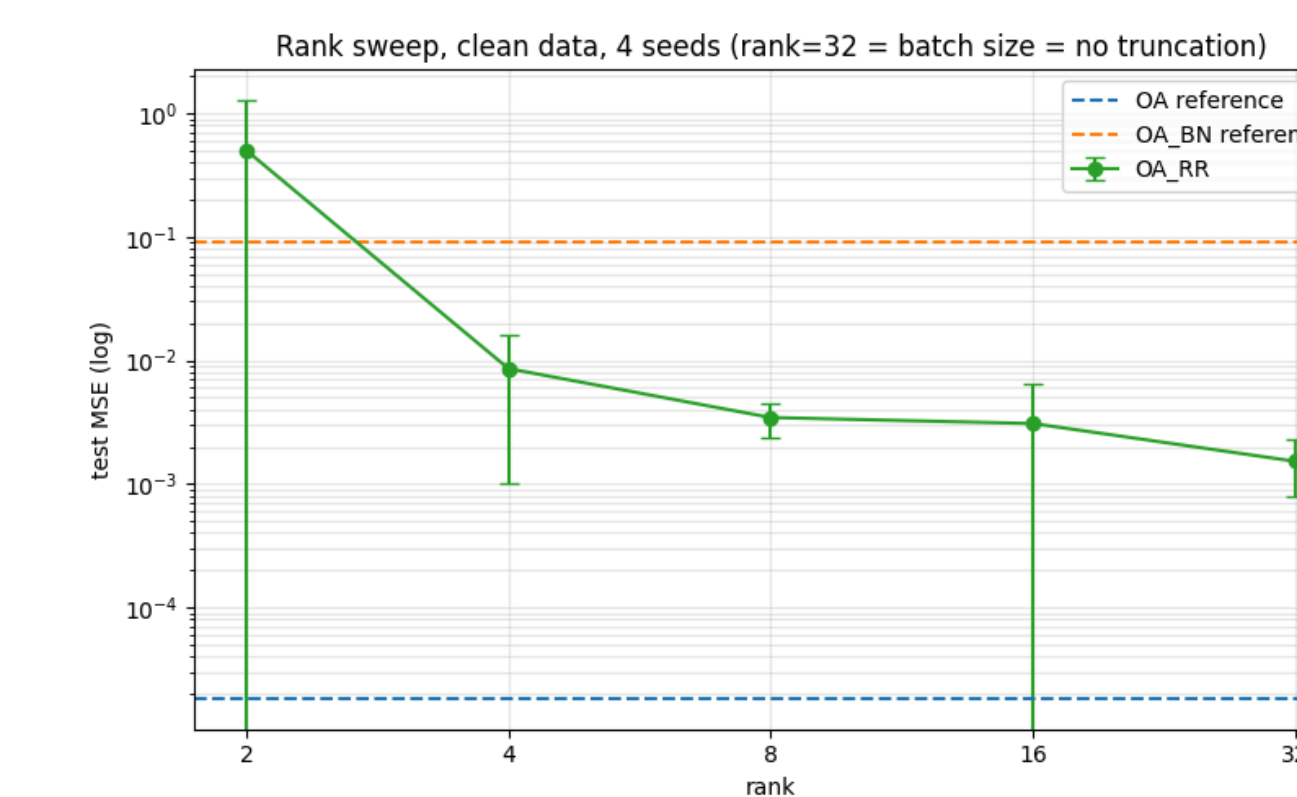


The three architectures: identical Conv/Linear weights, one slot changed.

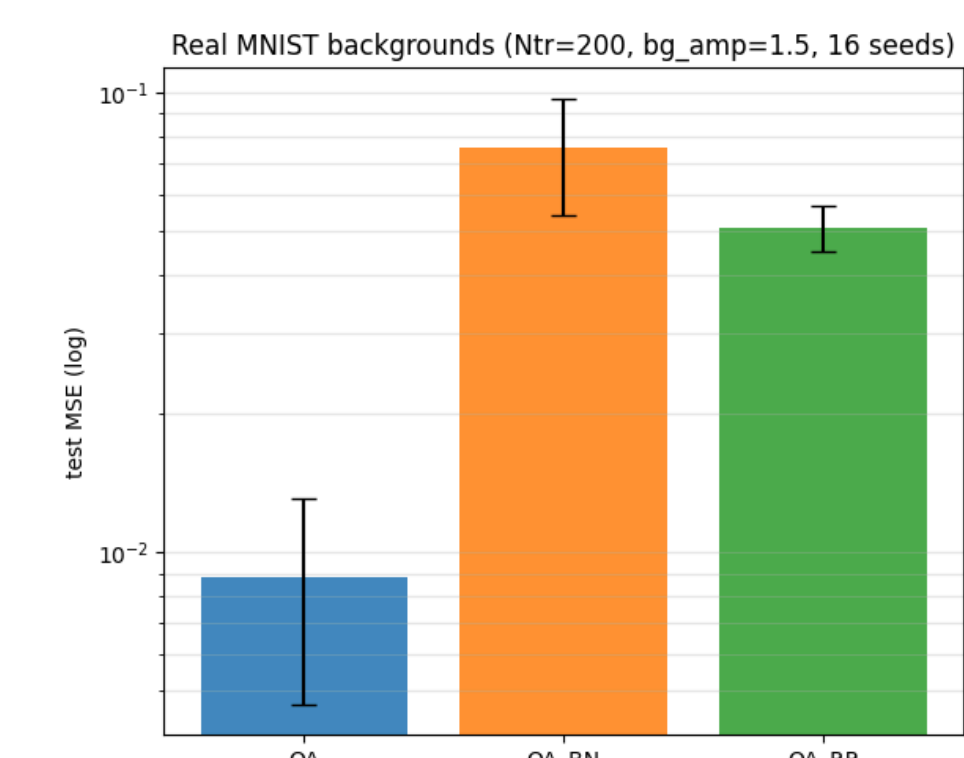
04. Results



Heavy input noise, 32 seeds: the per-layer SVD layer (green) beats BN *and* the plain net at every training-set size. (OA = plain, OA_RR = SVD layer.)



Error falls monotonically with rank k (bottleneck ablation).



Real backgrounds: plain wins; SVD still beats BN.

- SVD beats BN in every comparison we ran** (clean, noisy, small-data, real backgrounds), by 1.5–15×.
- Under noise it beats **even the unregularized net at every dataset size** (2–3×), with the tightest seed variance and near-zero train/test gap.
- Early training blow-ups came from a bug in our **hand-written SVD gradient**, not from the layer itself: with PyTorch's built-in SVD gradient, training is stable.

05. Conclusion

- The SVD layer is a **more robust drop-in alternative to BN**, and under input noise outperforms even the unregularized network at every dataset size we tested.
- Its one weakness: the singular values in Σ can have any values, so the **mean of the batch is not enforced to be small**. Encouraging singular values to go closer to 1 or to 0 is the **natural next step**.

References

J. Mounayer et al. *Rank Reduction Autoencoders*. arXiv:2405.13980, 2024.
S. Ioffe, C. Szegedy. *Batch Normalization*. ICML 2015.